

Plants in limbo — the theory of detrital photosynthesis

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Supplementary methods

Halophila ovalis ramets with several leaf pairs and intact roots were collected from Pelican Point in the brackish Swan Estuary (31.98715°S, 115.82101°E) alongside sediment from adjacent Matilda Bay (31.97572°S, 115.82324°E) on 15th September 2022. I filled the bottom of a 220-L glass flow-through seawater tank with the sediment and planted the ramets to acclimatise them to ambient seawater temperature and salinity and a 12-12-h light-dark cycle ($11 \pm 1.6 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, mean $\pm$ standard deviation, cf. Masini & Manning 1997) for about two hours. *Amphibolis antarctica* stems with several leaf clusters were collected from fully marine Marmion Lagoon (31.85303°S, 115.75092°E) on 13th October 2022 and immediately used in the experiment. Sediment and seagrasses were collected from less than one metre depth.

The ex situ decomposition experiment was commenced on the respective day of collection by excising individual leaves (*H. ovalis*) or leaf clusters (*A. antarctica*), hereinafter leaves, and immediately measuring photosynthesis (see next paragraph) of six (*H. ovalis*) or nine (*A. antarctica*) samples. Excision prematurely induces senescence and was intended to mimic detachment of non-senescent leaves, which may be caused by hydrodynamics or herbivory, rather than abscission. The remaining leaves were enclosed in mesh (3-mm diameter). Mesh bags were weighed down with inert aquarium pebbles and placed on the sediment in the same 220-L tank under identical photon fluence rate of $11 \pm 1.6 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, which is sufficient for detrital photosynthesis to play a role (Wright et al. 2024) and is representative of the light environment experienced by detritus in the understory of kelp forests and seagrass meadows or on deeper sediment flats. To measure photosynthesis at each of five (*A. antarctica*) to six (*H. ovalis*) timepoints over 34 days following excision, six destructive samples were taken from the mesh bags which were kept submerged for the entire duration of the experiment.

Light-saturated net photosynthesis (P_{max}) of leaves was measured using closed oxygen (O_2) incubations (cf. Wright et al. 2024). *A. antarctica* leaves weigh roughly ten times as much as *H. ovalis* leaves (given my data the mass ratio seems to be closer to 14:1). Since I wanted to measure both seagrasses at a comparable leaf mass per volume for the same duration, I incubated ten *H. ovalis* leaves or single *A. antarctica* leaves alongside seawater blanks in sealed glass jars filled with water collected from the flow-through system. Each jar was equipped with a magnetic stir bar and a self-adhesive planar O_2 sensor spot (SP-PSt3-SA-NAU-D5-YOP, PreSens Precision Sensing GmbH, Regensburg, Germany) and placed on a magnetic stirrer under a saturating (Masini & Manning 1997, Jamaludin et al. 2006, Said et al. 2021) irradiance of $420 \pm 19 \mu\text{mol photons m}^{-2} \text{s}^{-1}$ (Zeus, Ledzeal, Shenzhen Topline Lighting Technology Co. Ltd., Shenzhen, China) in a 20°C room. Care was taken to exclude air from jars. Dissolved O_2 (μM) was measured fibre-optically through the glass every 10 s over at least 35 min with a four-channel O_2 meter (OXY-4 SMA G2, PreSens Precision Sensing GmbH, Regensburg, Germany). The O_2 meter was calibrated using anoxic (1% w/v Na_2SO_3) and air-saturated (bubbled with air) ultrapure water. All measurements were corrected for incubation temperature ($18 \pm 0.73^\circ\text{C}$), pressure ($1020 \pm 6 \text{ hPa}$) and salinity ($35 \pm 0.48 \text{ ‰}$) using a single temperature dipping probe connected to the first channel of the built-in temperature sensor of the O_2 meter and placed in a fifth jar filled with seawater, the built-in pressure sensor of the O_2 meter and a handheld refractometer respectively. I blotted and weighed (0.01-g accuracy, 440-33N, Kern & Sohn GmbH, Balingen, Germany) leaves and gravimetrically determined jar volume with ultrapure water to standardise P_{max} as shown in Equation S1.

In the first instance, simple linear models with centred incubation time ($t - \bar{t}$, min) as the predictor and dissolved O_2 (μM) as the response variable were fit to measurements from each sample and blank incubation. This yielded posterior probability distributions for slopes (β , $\mu M \text{ min}^{-1}$) which were converted to mass-based $P_{\max}$ ($\mu mol \text{ g}^{-1} \text{ h}^{-1}$) as

$$P_{\max} = \frac{(\beta_s - \beta_b) \times V \times \Delta t}{m} \quad (S1)$$

where subscript s and b denote sample and blank incubations from the same measurement group, V is the incubation volume ($175 \pm 0.17 \text{ mL}$) in litres, m is the sample blotted mass ($0.52 \pm 0.23 \text{ g}$) in grams and Δt is the desired period (1 h) in minutes. I also calculated leaf-based $P_{\max}$. For *A. antarctica* this was simply $(\beta_s - \beta_b) \times V \times \Delta t$. For *H. ovalis* the rate was additionally divided by 10 because there were 10 leaves in each incubation chamber.

Varying conditions during the 35-min incubations described above could affect $P_{\max}$ and confound the experimental effect. Possible confounding variables that I measured are initial incubation O_2 (O , μM), mean incubation temperature (T , $^{\circ}C$), mean incubation pressure (P , hPA), incubation salinity (S , ‰) and sample mass (m , g). Although the oxygen measurements themselves were corrected for temperature, pressure and salinity, photosynthesis could also be influenced by these variables. Similarly, even though $P_{\max}$ was standardised by mass (Equation S1), the amount of leaf relative to water in the incubation chamber could additionally affect photosynthesis by changing the availability of carbon and nutrients. I explored these unwanted effects using a multiple linear regression of $P_{\max}$ against standardised (minus mean, divided by standard deviation) confounders:

$$\begin{aligned} P_{\max i} &\sim \text{Normal}(\overline{P_{\max i}}, \sigma_{P_{\max}}) \\ \overline{P_{\max i}} &= \alpha + \beta_O O_i + \beta_T T_i + \beta_P P_i + \beta_S S_i + \beta_m m_i \\ \alpha &\sim \text{Normal}(21, 10) \\ \beta_O, \beta_T, \beta_P, \beta_S, \beta_m &\sim \text{Normal}(0, 1) \\ \sigma_{P_{\max}} &\sim \text{Exponential}(1) \end{aligned} \quad (S2)$$

The likelihood for $P_{\max}$ is normal because predictions of net photosynthesis can take negative values. Each parameter was stratified by species. Even though $P_{\max}$ was only weakly associated with confounders (Figure S1), I decided to account for them in the main model. To predict $P_{\max}$ with detrital age (t , days), I used the logistic model of detrital photosynthesis (Equation 1) as

$$\begin{aligned} P_{\max i} &\sim \text{Normal}(\overline{P_{\max i}}, \sigma_{P_{\max}}) \\ \overline{P_{\max i}} &= \frac{\alpha_i + \tau_i}{5(t_i - \mu_i)} - \tau_i \\ &\quad \frac{1 + e^{-\mu_i}}{\mu_i} \\ \ln \alpha_i &= \ln \alpha + \beta_O O_i + \beta_T T_i + \beta_P P_i + \beta_S S_i + \beta_m m_i \\ \ln \mu_i &= \ln \mu + \gamma_O O_i + \gamma_T T_i + \gamma_P P_i + \gamma_S S_i + \gamma_m m_i \\ \ln \tau_i &= \ln \tau + \delta_O O_i + \delta_T T_i + \delta_P P_i + \delta_S S_i + \delta_m m_i \\ \ln \alpha &\sim \text{Normal}(\ln 21, 0.4) \\ \ln \mu &\sim \text{Normal}(\ln 14, 0.6) \\ \ln \tau &\sim \text{Normal}(\ln 2, 0.4) \\ \beta_O, \beta_T, \beta_P, \beta_S, \beta_m, \gamma_O, \gamma_T, \gamma_P, \gamma_S, \gamma_m, \delta_O, \delta_T, \delta_P, \delta_S, \delta_m &\sim \text{Normal}(0, 1) \\ \sigma_{P_{\max}} &\sim \text{Exponential}(1) \end{aligned} \quad (S3)$$

where α is baseline photosynthesis ($\mu mol \text{ g}^{-1} \text{ h}^{-1}$), μ is the photosynthetic half-life (days) and τ is respiration of dead leaves ($\mu mol \text{ g}^{-1} \text{ h}^{-1}$). These parameters are strictly positive, so their logarithm is described by a multiple linear regression against the confounders (cf. Equation S2). Some species-specific influence of confounders on α and μ is evident (Figure S1), so their inclusion in this model is important for accurate prediction. I also ran this model for leaf-based $P_{\max}$, adjusting the priors for α and τ accordingly.

Prior probability distributions for parameters were chosen based on published estimates. For the initial linear regressions of O_2 against incubation time, I centred the intercept prior on regional average seawater O_2 and the slope prior on zero. For α (Equation S2, S3), I chose the prior mean based on various published $P_{\max}$ estimates for *H. ovalis* ($40 \pm 57 \mu\text{mol g}^{-1} \text{h}^{-1}$, $1.8 \pm 2.5 \mu\text{mol leaf}^{-1} \text{h}^{-1}$, $n = 32$, Björk et al. 1997, Jamaludin et al. 2006, Borum et al. 2016, Lamit & Tanaka 2021, Said et al. 2021, 2026) and *A. antarctica* ($14 \pm 9 \mu\text{mol g}^{-1} \text{h}^{-1}$, $8.8 \pm 5.5 \mu\text{mol leaf}^{-1} \text{h}^{-1}$, $n = 8$, Masini & Manning 1997, Borum et al. 2016, Said et al. 2026). Units were converted where necessary by multiplying by mean dry-fresh mass ratios for *H. ovalis* ($0.21 \text{ g dry mass g}^{-1} \text{ fresh mass}$) and *A. antarctica* ($0.29 \text{ g dry mass g}^{-1} \text{ fresh mass}$) (de los Santos et al. 2012, Borum et al. 2016) and my own leaf fresh masses for *H. ovalis* ($0.04 \pm 0.01 \text{ g leaf}^{-1}$) and *A. antarctica* ($0.61 \pm 0.27 \text{ g leaf}^{-1}$). For μ (Equation S3), I settled on a median of 14 days, close to the minimum of 15 days for *Ecklonia radiata* (Wright et al. 2024) because I expected photosynthetic decay to be faster than for seaweeds. For τ (Equation S3), I centred the prior around published estimates of O_2 consumption by leaf litter of *Zostera marina* ($17 \pm 5.6 \mu\text{mol g}^{-1} \text{h}^{-1}$, $n = 16$, Blum & Mills 1991) and *Posidonia oceanica* ($1.3 \pm 1.3 \mu\text{mol g}^{-1} \text{h}^{-1}$, $n = 46$, Mateo & Romero 1996). Units were again converted by multiplying by mean dry-fresh mass ratios for *Z. marina* ($0.23 \text{ g dry mass g}^{-1} \text{ fresh mass}$, Evans et al. 1986) and *P. oceanica* ($0.24 \text{ g dry mass g}^{-1} \text{ fresh mass}$, Apostolaki et al. 2024). I chose a reasonable standard deviation for each prior based on variation in the published estimates and prior simulation.

The treatment of measurement error deserves brief mention. β and V (Equation S1) and hence $P_{\max}$ (Equation S1, S2, S3) as well as O_2 and T (Equation S2, S3) are measured with error, which I propagated into each downstream models as the standard deviation of upstream posteriors. This was achieved by parameterising latent variables for $P_{\max}$, O_2 and T . In all cases I used the error structure $\text{mean} \sim \text{Normal}(\text{latent}, \text{standard deviation})$ where each observed mean is described by a normal distribution centred on the unobserved true value with observed error (McElreath 2020). For the response variable $P_{\max}$, the latent variable is described by the normal likelihood (Equation S2, S3). For the standardised predictor variables O_2 and T , the latent variable is described by a standard normal prior: $\text{latent} \sim \text{Normal}(0, 1)$. Wherever a variable is measured with error, I visualised each observation as a distribution (violin or ellipse) rather than a point.

Data analysis and visualisation were performed in R v4.5.1 (R Core Team 2026) within the integrated development environment RStudio v2026.01.0+392 (RStudio Team 2026). General workflow was facilitated by the tidyverse package family v2.0.0 (Wickham et al. 2019) and here v1.0.2 (Müller 2025). Hamiltonian Monte Carlo models were written in Stan (Carpenter et al. 2017) and run with CmdStan v2.17.1 (Lee et al. 2026) via the R interface cmdstanr v0.9.0 (Gabry et al. 2025). All models were run with 8 Markov chains spread across all cores with 10^4 warmup and sampling iterations each. Convergence and smooth sampling were optimised by assessing $\hat{R}$ scores and visually scrutinising trace rank and pair plots with bayesplot v1.14.0 (Gabry et al. 2019). All reported results are posteriors and derived central tendencies and intervals calculated with tidybayes v3.0.7 (Kay 2024). Figure and table creation was facilitated by ggriidges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For detailed insight into data analysis and visualisation, all data, R and Stan code and diagnostic plots can be consulted at github.com/lukaseamus/plant-limbo. Vector graphics were polished in Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

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Supplementary figures

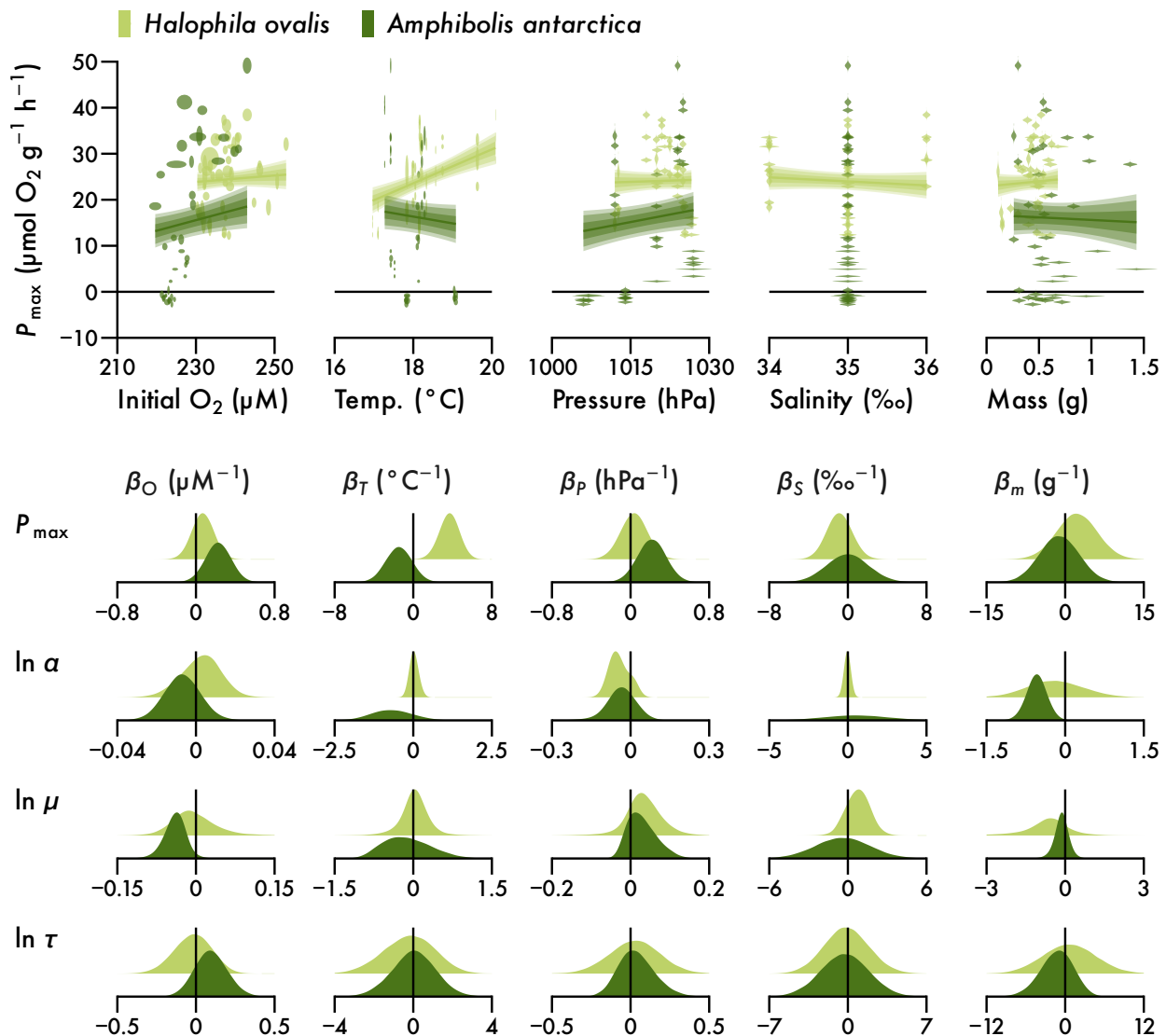


Figure S1. Effect of confounding variables on light-saturated net photosynthesis ($P_{\max}$) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves. $P_{\max}$ is standardised by leaf fresh mass. Ellipses and violins are kernel density estimates of observations with measurement error ($n = 42$ for *H. ovalis*, $n = 39$ for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted means. Distributions are kernel density estimates of slope posteriors (Equation S2, S3).

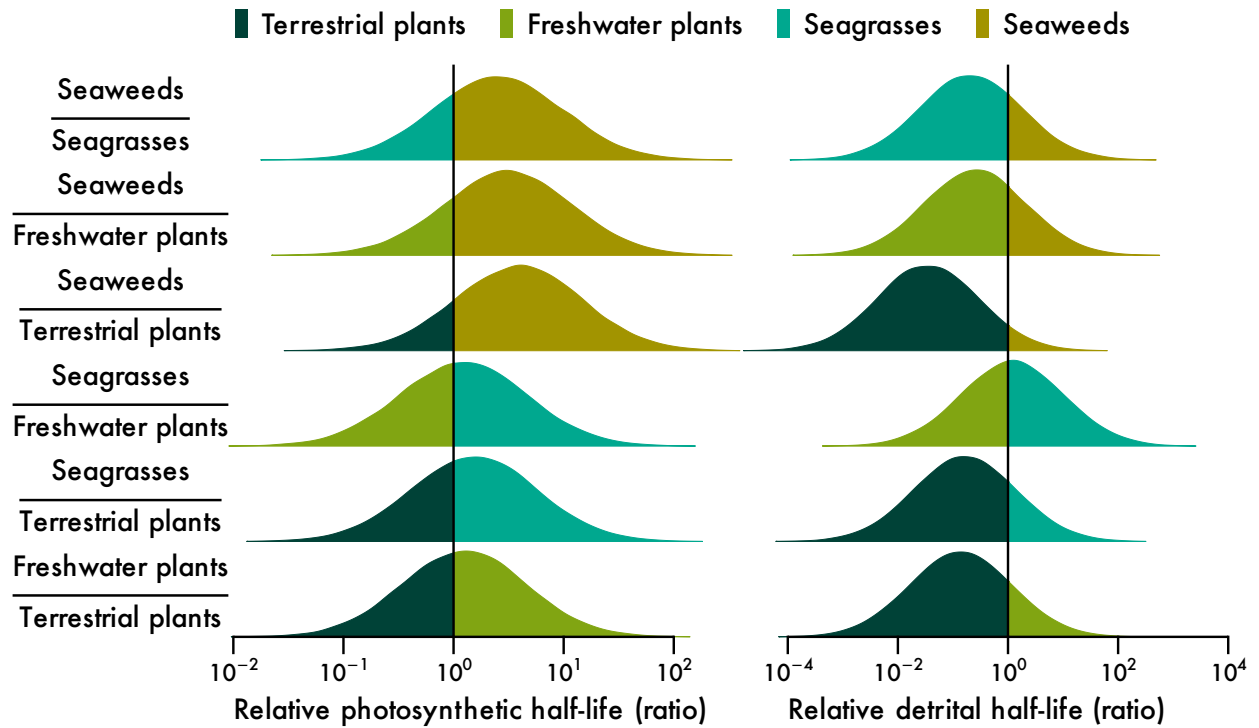


Figure S2. Photosynthetic and detrital half-life contrasts between plant groups. The group with more probability (coloured area) either side of the identity line at ratio = 1 has a greater half-life.

Supplementary tables

Table S1. Species included in the meta-analysis. Phyla are given in bold and orders are underlined. Seaweeds are comprised of Chlorophyta and Heterokontophyta. Seagrasses are marked with an asterisk. All remaining Alismatales are freshwater plants and the rest of Streptophyta is terrestrial. The longest experimental durations and largest numbers of studies, experiments and observations in each non-taxonomic group are highlighted in bold.

	Species	Days	Studies	Exper.	Observ.
Chlorophyta					
<u>Ulvales</u>					
Ulvaceae	<i>Ulva clathrata</i>	30	1	1	6
	<i>Ulva intestinalis</i>	30	1	1	6
	<i>Ulva reticulata</i>	30	1	1	6
Heterokontophyta					
<u>Desmarestiales</u>					
Desmarestiaceae	<i>Desmarestia anceps</i>	313	1	1	65
<u>Fucales</u>					
Durvillaeaceae	<i>Durvillaea antarctica</i>	14	2	16	448
Sargassaceae	<i>Sargassum fallax</i>	14	1	9	84
	<i>Sargassum spinuligerum</i>	14	1	13	159
Seirococcaceae	<i>Scytothalia dorycarpa</i>	14	1	6	48
<u>Laminariales</u>					
Laminariaceae	<i>Laminaria digitata</i>	46	4	6	191
	<i>Laminaria hyperborea</i>	168	7	56	559
	<i>Laminaria ochroleuca</i>	56	4	51	416
	<i>Macrocystis pyrifera</i>	55	4	23	375
	<i>Saccharina latissima</i>	39	2	2	55
Lessoniaceae	<i>Ecklonia radiata</i>	119	2	10	117
<u>Tilopteridales</u>					
Phyllariaceae	<i>Saccorhiza polyschides</i>	39	1	1	38
Streptophyta					
<u>Alismatales</u>					
Cymodoceaceae	<i>Amphibolis antarctica</i> *	34	1	1	39
Hydrocharitaceae	<i>Elodea densa</i>	16	2	3	43
	<i>Halophila ovalis</i> *	34	1	1	42
	<i>Hydrilla verticillata</i>	12	7	16	506
	<i>Thalassia testudinum</i> *	52	1	1	14
	<i>Vallisneria americana</i>	104	1	1	5
	<i>Vallisneria spiralis</i>	9	6	6	59
Potamogetonaceae	<i>Potamogeton</i> sp.	104	1	1	5
	<i>Stuckenia pectinata</i>	9	5	5	55
Zosteraceae	<i>Zostera marina</i> *	85	1	1	7
	<i>Zostera muelleri</i> *	16	1	4	58
<u>Apiales</u>					
Apiaceae	<i>Coriandrum sativum</i>	9	1	1	24
	<i>Petroselinum crispum</i>	7	1	1	7
<u>Asterales</u>					
Asteraceae	<i>Chrysanthemum</i> × <i>morifolium</i>	12	1	2	30
	<i>Helianthus annuus</i>	0.024	1	1	36
	<i>Helianthus tuberosus</i>	0.026	1	1	36
	<i>Inula racemosa</i>	0.025	1	1	36

	Species	Days	Studies	Exper.	Observ.
	<i>Smallanthus connatus</i>	0.026	1	1	36
<u>Brassicales</u>					
Brassicaceae	<i>Arabidopsis thaliana</i>	6	2	3	60
	<i>Brassica napus</i>	1.7	1	25	125
	<i>Brassica oleracea</i>	10	1	3	22
	<i>Nasturtium officinale</i>	5	2	3	39
	<i>Raphanus raphanistrum</i>	6	1	4	20
	<i>Sinapis alba</i>	6	1	4	20
Tropaeolaceae	<i>Tropaeolum majus</i>	8	2	3	27
<u>Caryophyllales</u>					
Amaranthaceae	<i>Amaranthus cruentus</i>	0.026	1	1	36
	<i>Celosia argentea</i>	0.024	1	1	36
	<i>Chenopodium berlandieri</i>	0.27	1	2	26
	<i>Gomphrena serrata</i>	0.024	1	1	36
	<i>Spinacia oleracea</i>	10	3	4	85
Polygonaceae	<i>Koenigia weyrichii</i>	0.028	1	1	36
<u>Cucurbitales</u>					
Cucurbitaceae	<i>Cucumis sativus</i>	13	2	3	13
	<i>Sicyos edulis</i>	10	1	2	48
<u>Ericales</u>					
Theaceae	<i>Camellia sinensis</i>	4	1	2	11
<u>Fabales</u>					
Fabaceae	<i>Cercis canadensis</i>	0.0052	1	4	576
	<i>Lathyrus oleraceus</i>	17	2	2	42
	<i>Phaseolus vulgaris</i>	2.1	1	1	10
	<i>Trifolium subterraneum</i>	7	2	5	25
	<i>Vigna unguiculata</i>	4	1	10	170
<u>Fagales</u>					
Betulaceae	<i>Betula pendula</i>	0.029	1	1	36
Fagaceae	<i>Quercus alba</i>	0.01	1	1	24
	<i>Quercus muehlenbergii</i>	0.0052	1	1	192
	<i>Quercus robur</i>	0.024	2	2	228
	<i>Quercus rubra</i>	0.01	1	1	24
	<i>Juglans nigra</i>	0.01	1	1	24
Juglandaceae					
<u>Lamiales</u>					
Lamiaceae	<i>Salvia officinalis</i>	11	1	1	8
<u>Laurales</u>					
Lauraceae	<i>Cinnamomum tamala</i>	21	1	1	36
<u>Pinales</u>					
Pinaceae	<i>Larix sibirica</i>	0.025	1	1	36
	<i>Pinus sibirica</i>	0.023	1	1	33
	<i>Pinus sylvestris</i>	0.022	1	1	33
<u>Poales</u>					
Cyperaceae	<i>Bolboschoenus fluviatilis</i>	104	1	1	5
Poaceae	<i>Avena sativa</i>	8	8	28	161
	<i>Hordeum vulgare</i>	11	13	25	466
	<i>Lolium multiflorum</i>	5	1	3	44
	<i>Lolium pratense</i>	7	5	8	112
	<i>Oryza sativa</i>	27	11	48	327
	<i>Panicum miliaceum</i>	5	2	2	69
	<i>Secale cereale</i>	5	2	9	102

	Species	Days	Studies	Exper.	Observ.
	<i>Triticum aestivum</i>	10	13	41	646
	<i>Triticum turgidum</i>	6	1	1	7
	<i>Triticum vulgare</i>	6	1	2	48
	<i>Zea mays</i>	6	3	4	58
Typhaceae	<i>Typha angustifolia</i>	104	1	1	5
Polypodiales					
Nephrolepidaceae	<i>Nephrolepis exaltata</i>	10	1	4	24
<u>Rosales</u>					
Rosaceae	<i>Fragaria × ananassa</i>	0.0056	1	1	663
	<i>Malus domestica</i>	21	3	6	150
	<i>Malus hupehensis</i>	15	1	1	18
<u>Sapindales</u>					
Rutaceae	<i>Citrus × aurantium</i>	10	1	1	5
Sapindaceae	<i>Acer saccharinum</i>	0.25	1	4	36
	<i>Acer truncatum</i>	0.0052	1	1	192
<u>Solanales</u>					
Convolvulaceae	<i>Ipomoea batatas</i>	3	1	2	24
Solanaceae	<i>Nicotiana rustica</i>	10	1	2	180
	<i>Nicotiana tabacum</i>	9	1	2	24
	<i>Solanum lycopersicum</i>	5	1	6	192
	<i>Solanum melongena</i>	16	1	1	24
	<i>Solanum tuberosum</i>	0.026	1	1	36
<u>Vitales</u>					
Vitaceae	<i>Vitis vinifera</i>	0.0052	1	5	1280

Table S2. Predictions of photosynthetic half-life (days, μ in Equation 2) for new experiments on unobserved species in each plant group in this study as well as new plant groups. Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean $\pm$ standard deviation).

Light	Response	Photosynthetic half-life
Terrestrial plants		
Light	Photosynthesis	13 (2.6 ± 0.94)
	Chlorophyll	18 (2.9 ± 0.93)
Dark	Photosynthesis	7.2 (2 ± 0.92)
	Chlorophyll	9.5 (2.3 ± 0.92)
Freshwater plants		
Light	Chlorophyll	19 (2.9 ± 0.98)
Dark	Photosynthesis	11 (2.4 ± 0.95)
	Chlorophyll	16 (2.8 ± 0.95)
Seagrasses		
Light	Photosynthesis	19 (2.9 ± 0.98)
	Chlorophyll	26 (3.3 ± 0.98)
Dark	Chlorophyll	16 (2.8 ± 1)
Seaweeds		
Light	Photosynthesis	51 (3.9 ± 0.94)
	Chlorophyll	81 (4.4 ± 0.96)
Dark	Photosynthesis	24 (3.2 ± 1)
All plants		
Light	Photosynthesis	17 (2.8 ± 1.1)
	Chlorophyll	24 (3.2 ± 1.1)
Dark	Photosynthesis	12 (2.4 ± 1.1)
	Chlorophyll	16 (2.8 ± 1.1)

Table S3. Predictions of photosynthetic half-life (days, μ in Equation 2) for new experiments on each species in this study (cf. Table S1 for complete taxonomy). Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean $\pm$ standard deviation).

Species	Light	Response	Photosynthetic half-life
Seaweeds			
<i>Desmarestia anceps</i>	Light	Chlorophyll	154 (5 ± 0.9)
<i>Durvillaea antarctica</i>	Light	Photosynthesis	59 (4.1 ± 0.87)
		Chlorophyll	95 (4.6 ± 0.9)
<i>Ecklonia radiata</i>	Light	Photosynthesis	32 (3.5 ± 0.78)
	Dark	Photosynthesis	15 (2.7 ± 0.87)
<i>Laminaria digitata</i>	Light	Photosynthesis	68 (4.2 ± 0.86)
		Chlorophyll	109 (4.7 ± 0.88)
<i>Laminaria hyperborea</i>	Light	Photosynthesis	62 (4.1 ± 0.73)
		Chlorophyll	99 (4.6 ± 0.79)
<i>Laminaria ochroleuca</i>	Light	Photosynthesis	68 (4.2 ± 0.74)
		Chlorophyll	108 (4.7 ± 0.79)
<i>Macrocystis pyrifera</i>	Light	Photosynthesis	60 (4.1 ± 0.84)
		Chlorophyll	96 (4.6 ± 0.86)
<i>Saccharina latissima</i>	Light	Photosynthesis	60 (4.1 ± 0.9)
		Chlorophyll	96 (4.6 ± 0.91)
<i>Saccorhiza polyschides</i>	Light	Chlorophyll	87 (4.5 ± 0.92)

Species	Light	Response	Photosynthetic half-life
<i>Sargassum fallax</i>	Light	Photosynthesis	30 (3.4 ± 0.82)
<i>Sargassum spinuligerum</i>	Light	Photosynthesis	64 (4.2 ± 0.86)
		Chlorophyll	102 (4.6 ± 0.89)
<i>Scytothalia dorycarpa</i>	Light	Photosynthesis	45 (3.8 ± 0.95)
<i>Ulva clathrata</i>	Light	Chlorophyll	85 (4.4 ± 0.94)
<i>Ulva intestinalis</i>	Light	Chlorophyll	84 (4.4 ± 0.94)
<i>Ulva reticulata</i>	Light	Chlorophyll	85 (4.4 ± 0.94)
Seagrasses			
<i>Amphibolis antarctica</i>	Light	Photosynthesis	14 (2.6 ± 0.88)
<i>Halophila ovalis</i>	Light	Photosynthesis	27 (3.3 ± 0.91)
<i>Thalassia testudinum</i>	Light	Chlorophyll	26 (3.2 ± 0.87)
<i>Zostera marina</i>	Light	Chlorophyll	26 (3.3 ± 0.87)
<i>Zostera muelleri</i>	Light	Chlorophyll	25 (3.2 ± 0.86)
	Dark	Chlorophyll	16 (2.8 ± 0.86)
Freshwater plants			
<i>Elodea densa</i>	Light	Chlorophyll	18 (2.9 ± 0.89)
	Dark	Chlorophyll	15 (2.7 ± 0.86)
<i>Hydrilla verticillata</i>	Light	Chlorophyll	14 (2.6 ± 0.79)
	Dark	Photosynthesis	7.7 (2 ± 0.78)
	Dark	Chlorophyll	12 (2.5 ± 0.78)
<i>Potamogeton sp.</i>	Dark	Chlorophyll	25 (3.2 ± 0.88)
<i>Stuckenia pectinata</i>	Dark	Photosynthesis	7.1 (2 ± 0.8)
	Dark	Chlorophyll	11 (2.4 ± 0.82)
<i>Vallisneria americana</i>	Dark	Chlorophyll	25 (3.2 ± 0.88)
<i>Vallisneria spiralis</i>	Dark	Photosynthesis	7.8 (2.1 ± 0.79)
	Dark	Chlorophyll	12 (2.5 ± 0.81)
Terrestrial plants			
<i>Acer saccharinum</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Acer truncatum</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Amaranthus cruentus</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Arabidopsis thaliana</i>	Dark	Photosynthesis	4.3 (1.5 ± 0.81)
	Dark	Chlorophyll	5.8 (1.8 ± 0.8)
<i>Avena sativa</i>	Light	Chlorophyll	11 (2.4 ± 0.78)
	Dark	Chlorophyll	5.8 (1.8 ± 0.76)
<i>Betula pendula</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Bolboschoenus fluviatilis</i>	Dark	Chlorophyll	20 (3 ± 0.9)
<i>Brassica napus</i>	Light	Chlorophyll	18 (2.9 ± 0.91)
<i>Brassica oleracea</i>	Dark	Chlorophyll	8.9 (2.2 ± 0.85)
<i>Camellia sinensis</i>	Light	Photosynthesis	14 (2.7 ± 0.91)
<i>Celosia argentea</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Cercis canadensis</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Chenopodium berlandieri</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Chrysanthemum × morifolium</i>	Dark	Chlorophyll	9.4 (2.2 ± 0.82)
<i>Cinnamomum tamala</i>	Dark	Chlorophyll	9.6 (2.3 ± 0.84)
<i>Citrus × aurantium</i>	Light	Chlorophyll	19 (2.9 ± 0.9)
<i>Coriandrum sativum</i>	Dark	Chlorophyll	9.6 (2.3 ± 0.87)
<i>Cucumis sativus</i>	Light	Chlorophyll	19 (3 ± 0.89)
	Dark	Chlorophyll	10 (2.3 ± 0.88)
<i>Fragaria × ananassa</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Gomphrena serrata</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Helianthus annuus</i>	Light	Photosynthesis	13 (2.6 ± 0.94)

Species	Light	Response	Photosynthetic half-life
<i>Helianthus tuberosus</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Hordeum vulgare</i>	Light	Photosynthesis	9.2 (2.2 ± 0.77)
		Chlorophyll	12 (2.5 ± 0.76)
		Photosynthesis	5 (1.6 ± 0.75)
<i>Inula racemosa</i>	Dark	Chlorophyll	6.6 (1.9 ± 0.75)
		Photosynthesis	13 (2.6 ± 0.94)
		Photosynthesis	7.9 (2.1 ± 0.9)
<i>Ipomoea batatas</i>	Dark	Chlorophyll	11 (2.4 ± 0.9)
<i>Juglans nigra</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Koenigia weyrichii</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Larix sibirica</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Lathyrus oleraceus</i>	Light	Photosynthesis	16 (2.8 ± 0.89)
		Chlorophyll	22 (3.1 ± 0.88)
		Photosynthesis	12 (2.5 ± 0.92)
<i>Lolium multiflorum</i>	Light	Chlorophyll	16 (2.8 ± 0.92)
		Photosynthesis	4.1 (1.4 ± 0.77)
		Chlorophyll	5.5 (1.7 ± 0.77)
<i>Lolium pratense</i>	Dark	Photosynthesis	24 (3.2 ± 0.86)
		Chlorophyll	33 (3.5 ± 0.85)
		Photosynthesis	13 (2.6 ± 0.85)
<i>Malus domestica</i>	Light	Chlorophyll	18 (2.9 ± 0.85)
		Chlorophyll	13 (2.6 ± 0.88)
		Chlorophyll	4.3 (1.5 ± 0.81)
<i>Malus hupehensis</i>	Dark	Chlorophyll	18 (2.9 ± 0.87)
<i>Nasturtium officinale</i>	Dark	Chlorophyll	24 (3.2 ± 0.86)
<i>Nephrolepis exaltata</i>	Light	Photosynthesis	9.8 (2.3 ± 0.86)
		Chlorophyll	13 (2.6 ± 0.85)
		Photosynthesis	8.5 (2.1 ± 0.84)
<i>Nicotiana rustica</i>	Dark	Chlorophyll	11 (2.4 ± 0.83)
		Photosynthesis	16 (2.8 ± 0.89)
		Chlorophyll	21 (3.1 ± 0.88)
<i>Nicotiana tabacum</i>	Light	Photosynthesis	10 (2.4 ± 0.77)
		Chlorophyll	14 (2.6 ± 0.76)
		Photosynthesis	5.7 (1.7 ± 0.76)
<i>Oryza sativa</i>	Dark	Chlorophyll	7.5 (2 ± 0.75)
		Chlorophyll	5.4 (1.7 ± 0.82)
		Chlorophyll	10 (2.3 ± 0.89)
<i>Panicum miliaceum</i>	Dark	Chlorophyll	13 (2.6 ± 0.93)
<i>Petroselinum crispum</i>	Dark	Chlorophyll	13 (2.6 ± 0.93)
<i>Phaseolus vulgaris</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Pinus sibirica</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Pinus sylvestris</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Quercus alba</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Quercus muehlenbergii</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Quercus robur</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Quercus rubra</i>	Light	Photosynthesis	13 (2.6 ± 0.94)
<i>Raphanus raphanistrum</i>	Light	Photosynthesis	14 (2.7 ± 0.89)
		Chlorophyll	19 (3 ± 0.89)
		Chlorophyll	12 (2.5 ± 0.89)
<i>Salvia officinalis</i>	Dark	Chlorophyll	13 (2.6 ± 0.86)
<i>Secale cereale</i>	Light	Photosynthesis	18 (2.9 ± 0.85)
		Chlorophyll	7.2 (2 ± 0.85)
		Chlorophyll	9.5 (2.3 ± 0.85)

Species	Light	Response	Photosynthetic half-life
<i>Sicyos edulis</i>	Dark	Photosynthesis	7.4 (2 ± 0.82)
		Chlorophyll	9.9 (2.3 ± 0.82)
<i>Sinapis alba</i>	Light	Photosynthesis	15 (2.7 ± 0.89)
		Chlorophyll	20 (3 ± 0.89)
<i>Smallanthus connatus</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Solanum lycopersicum</i>	Light	Photosynthesis	11 (2.4 ± 0.85)
<i>Solanum melongena</i>	Light	Photosynthesis	15 (2.7 ± 0.88)
<i>Solanum tuberosum</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Spinacia oleracea</i>	Dark	Photosynthesis	5.8 (1.8 ± 0.8)
		Chlorophyll	7.7 (2 ± 0.79)
<i>Trifolium subterraneum</i>	Light	Photosynthesis	12 (2.5 ± 0.88)
		Chlorophyll	16 (2.8 ± 0.88)
<i>Triticum aestivum</i>	Light	Photosynthesis	5.5 (1.7 ± 0.75)
		Chlorophyll	7.3 (2 ± 0.75)
	Dark	Photosynthesis	3 (1.1 ± 0.74)
		Chlorophyll	4 (1.4 ± 0.74)
<i>Triticum turgidum</i>	Light	Chlorophyll	18 (2.9 ± 0.92)
<i>Triticum vulgare</i>	Light	Photosynthesis	13 (2.6 ± 0.88)
	Dark	Photosynthesis	7.3 (2 ± 0.87)
<i>Tropaeolum majus</i>	Light	Chlorophyll	20 (3 ± 0.87)
	Dark	Chlorophyll	11 (2.4 ± 0.87)
<i>Typha angustifolia</i>	Dark	Chlorophyll	22 (3.1 ± 0.91)
<i>Vigna unguiculata</i>	Dark	Photosynthesis	5.3 (1.7 ± 0.78)
		Chlorophyll	7.2 (2 ± 0.78)
<i>Vitis vinifera</i>	Light	Photosynthesis	13 (2.6 ± 0.93)
<i>Zea mays</i>	Light	Photosynthesis	14 (2.6 ± 0.91)
		Chlorophyll	19 (2.9 ± 0.9)
	Dark	Chlorophyll	10 (2.3 ± 0.89)

Supplementary references

Studies included in the meta-analysis in addition to the present study. Note that 130 references are listed because the data from three studies were not independent and thus merged.

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